Piggy

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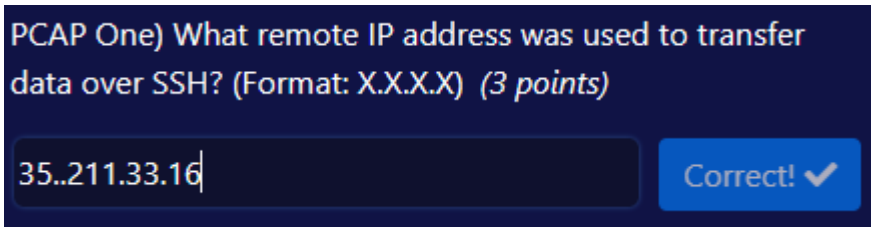
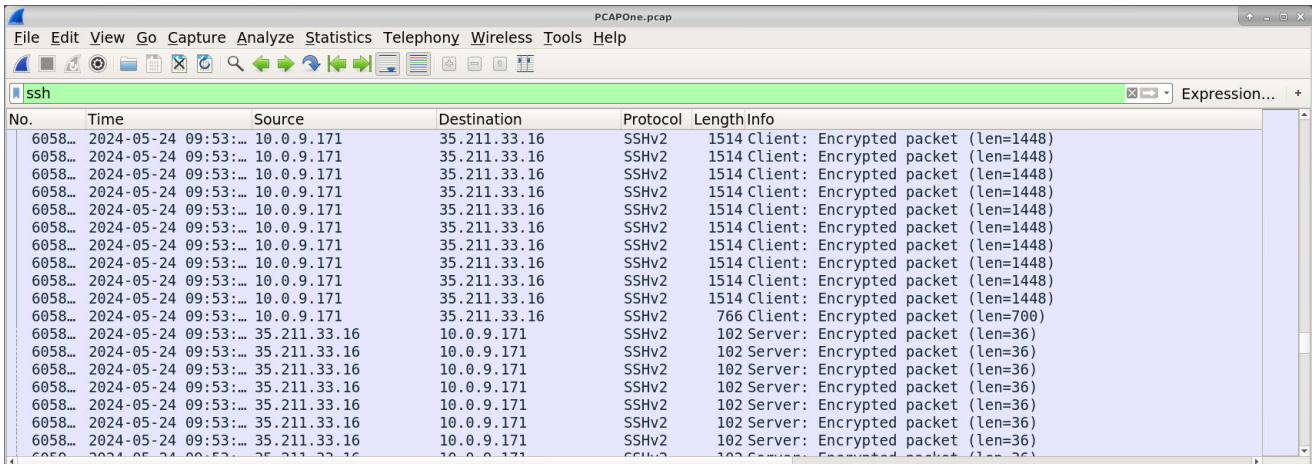
Tech Summary:

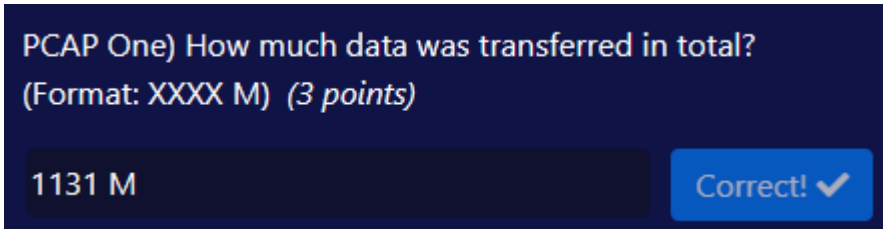
- **ASN (**Autonomous System Number**):** ASN reputation is used in cybersecurity to identify, block, or filter traffic during DDoS attacks, as described in this ThreatNG Security article.
- How to Identify **TXT query** in Wireshark? ---> dns.qry.type ==16 [Because the type number of TXT is 16]
- Ip addresses can be assorciated with malware family on OSINT websites.

Questions Investigation and screenshots:





we notice that there are two ip address extranging encrypted packets. The Ip starting with 35 is a remote ip address.



Go to Statistics-->Endpoints -->IPv4.

Wireshark · Endpoints · PCAPOne.pcap

Ethernet · 3 | IPv4 · 6 | IPv6 | TCP · 9 | UDP · 9

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	AS Number
10.0.0.2	16	1989	8	1195	8	794	—	—	—
10.0.9.171	869715	1145 M	778842	1136 M	90873	8483 k	—	—	—
34.110.209.165	6	412	2	140	4	272	—	—	—
35.211.33.16	856033	1131 M	87848	8211 k	768185	1123 M	—	—	—
52.94.51.164	10	607	5	300	5	307	—	—	—
82.2.64.107	13650	13 M	3010	271 k	10640	13 M	—	—	—

Name resolution | Limit to display filter | Endpoint Types

Help | Copy | Close

PCAP Two) Review the IPs the infected system has communicated with. Perform OSINT searches to identify the malware family tied to this infrastructure (Format: MalwareName) *(3 points)*

trickbot

Correct!

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	AS Number
10.0.0.2	12	1464	6	878	6	586	—	—	—
10.0.9.171	3827	4412 k	3228	4350 k	599	61 k	—	—	—
31.184.253.37	6	412	2	140	4	272	—	—	—
34.110.209.165	46	12 k	22	9986	24	2359	—	—	—
78.155.206.172	1	74	0	0	1	74	—	—	—
82.2.64.107	3754	4397 k	567	50 k	3187	4346 k	—	—	—
92.53.67.7	1	74	0	0	1	74	—	—	—
188.120.241.27	1	74	0	0	1	74	—	—	—
195.161.41.93	6	412	2	140	4	272	—	—	—

82.2.64.107 is key word to Search on OSINT websites. But It is clear. So we go back to the Wirshark,

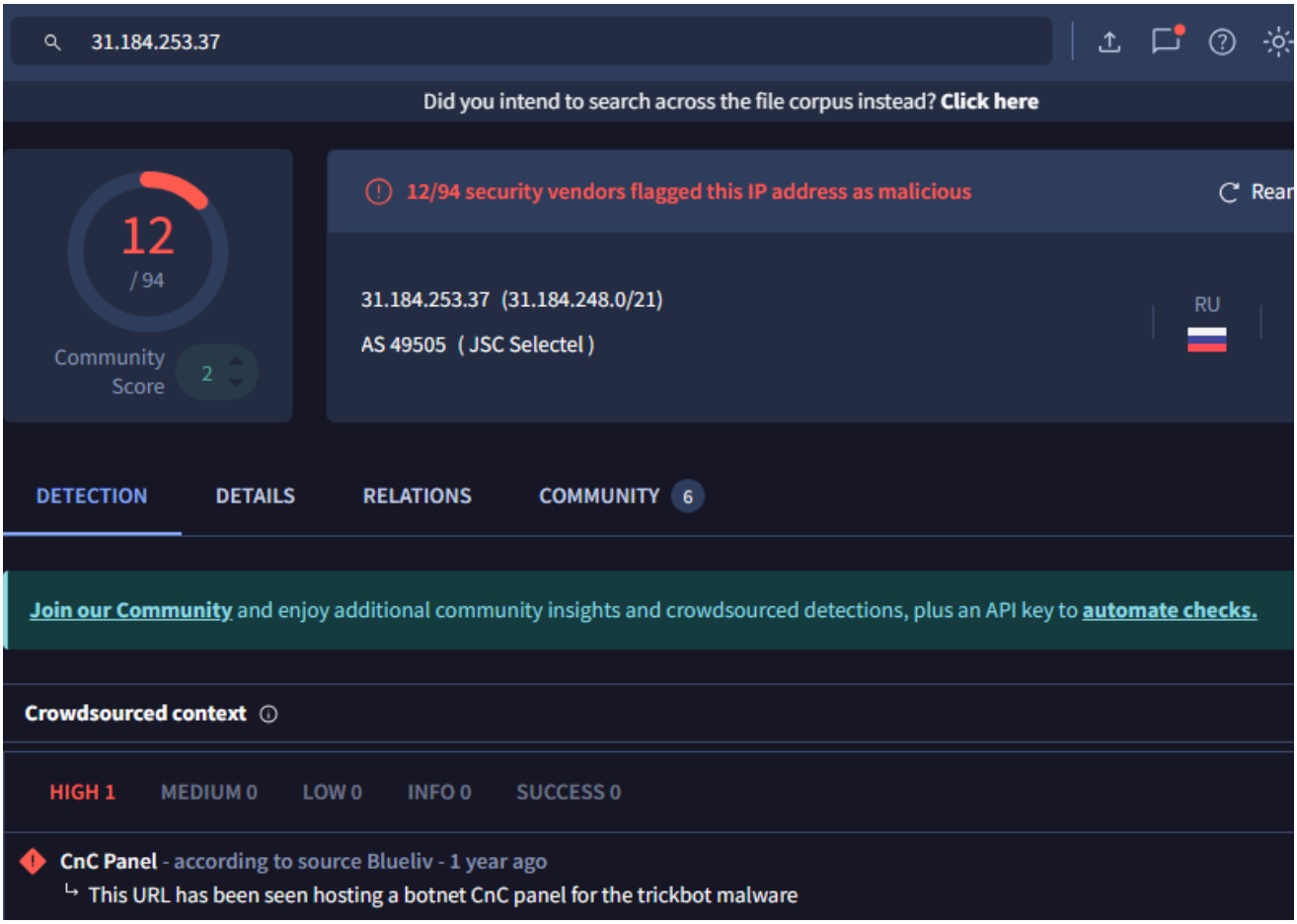
Wireshark · Conversations · PCAPTwo.pcap

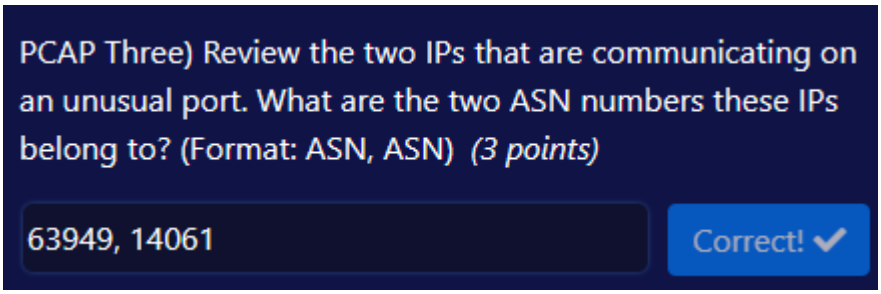
Ethernet · 1 | IPv4 · 8 | IPv6 | TCP · 8 | UDP · 6

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel S
10.0.0.2	10.0.9.171	12	1464	6	878	6	586	0.9
10.0.9.171	82.2.64.107	3754	4397 k	3187	4346 k	567	50 k	0.0
10.0.9.171	34.110.209.165	46	12 k	24	2359	22	9986	1.0
10.0.9.171	188.120.241.27	1	74	1	74	0	0	2.1
10.0.9.171	195.161.41.93	6	412	4	272	2	140	3.1
10.0.9.171	92.53.67.7	1	74	1	74	0	0	4.1
10.0.9.171	31.184.253.37	6	412	4	272	2	140	5.1
10.0.9.171	78.155.206.172	1	74	1	74	0	0	6.1

34.110.209.165 is next key word to search, because it is the second ip has the most conversation. However, This ip is also clear.

Then, the next one is195.161.41.93 and 31.184.253.37. They share the same packets. Finally, we get the answer.





Key Aspects of ASN (Autonomous System Number):

- **Purpose:** Enables routers to direct traffic across the internet using Border Gateway Protocol (BGP), acting like a "city" in the network infrastructure.
- **Usage Examples:** Used by Internet Service Providers (ISPs), technology firms (e.g., AWS, Microsoft), and corporations to manage their own public IP address blocks and traffic.
- **Components:** Consists of 16-digit (formerly) or 32-digit numbers managed by Regional Internet Registries (RIRs) like ARIN or RIPE NCC.
- **Security:** ASN reputation is used in cybersecurity to identify, block, or filter traffic during DDoS attacks, as described in this ThreatNG Security article.

Wireshark · Conversations · PCAPThree.pcap

Ethernet · 1 | IPv4 · 5 | IPv6 | TCP · 5 | UDP · 1

Address A	Port A	Address B	Port B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A
10.0.9.171	58032	34.110.209.165	443	6	412	4	272	2
10.0.9.171	58048	34.110.209.165	443	31	10 k	18	1828	13
10.0.9.171	34825	194.233.171.171	8080	12	1062	7	542	5
10.0.9.171	58651	104.236.57.24	8000	12	1062	7	542	5
82.2.64.107	56171	10.0.9.171	443	2516	2709 k	480	43 k	2036

Name resolution | Limit to display filter | Absolute start time | Conversation Types

Help | Copy | Follow Stream... | Graph... | Close

194.233.171.171 and 104.236.57.24 are using ports 8080 and 8000 which are usually used to test period.

194.233.171.171

Star

Frankfurt am Main, Hesse, DE | Cloud | Hosting | SSH | Webserver

- Summary
- Geolocation
- Privacy
- ASN
- Company
- Abuse
- Hosted domains

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Summary

ASN	AS63949 - Akamai Connected Cloud
Hostname	194-233-171-171.ip.linodeusercontent.com
Range	194.233.168.0/21
Company	Linode, LLC
Hosted domains	6
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@linode.com

IPinfo 104.236.57.24 Products Data Why IPinfo? Pricing Resources Docs Login Sign up

All IP Ranges > 104.0.0.0/8 > 104.236.0.0/16 > 104.236.57.0/24 > 104.236.57.24

104.236.57.24

Star

Clifton, New Jersey, US | Cloud | Hosting | SSH

- Summary
- Geolocation
- Privacy
- ASN
- Company
- Abuse

Need more data or want to access it via API or data downloads? Sign up to get free access — Sign up for free ›

Summary

ASN	AS14061 - DigitalOcean, LLC
Hostname	No Hostname
Range	104.236.0.0/18
Company	DigitalOcean, LLC
Hosted domains	0
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@digitalocean.com

PCAP Three) Perform OSINT checks. What malware category have these IPs been attributed to historically? (Format: MalwareType) *(3 points)*

miner

Correct!

0 / 94

Community Score

10+ detected files communicating with this IP address

Reanalyze More

104.236.57.24 (104.236.0.0/16)

AS 14061 (DigitalOcean, LLC)

US

Last Analysis Date

3 days ago

DETECTION DETAILS RELATIONS COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Do you want to automate checks?

AlphaSOC	Miner	GCP Abuse Intelligence	Miner
SOCRadar	Suspicious	Abusix	Clean

PCAP Three) What ATT&CK technique is most closely related to this activity? (Format: TXXXX) *(3 points)*

T1496 Correct! ✔

MITRE ATT miner

More results from mitre.org »

MITRE ATT&CK®
https://attack.mitre.org › techniques

Compute Hijacking, Sub-technique T1496.001 - Enterprise

Sep 25, 2024 — One common purpose for Compute Hijacking is to **validate transactions of cryptocurrency networks** and earn virtual currency. Read more

MITRE ATT&CK®
https://attack.mitre.org › techniques

Resource Hijacking, Technique T1496 - Enterprise

Apr 17, 2019 — Adversaries may leverage the resources of co-opted systems to complete resource-intensive tasks, which may impact system and/or hosted service availability. Read more

PCAP Four) Go to View > Time Display Format > Seconds Since Beginning of Capture. How long into the capture was the first TXT record query made? (Use the default time, which is seconds since the packet capture started) (Format: X.xxxxxx) *(3 points)*

8.527712 Correct! ✔

- What is TXT record query ? ---> DNS
- How to Identify TXT query in Wireshark? ---> dns.qry.type ==16 [Because the type number of TXT is 16]

PCAPFour.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

Title: Time Type: Time (format as specified) Fields: Enter a field ... Oc

No.	Time	Source	Destination	Protocol	Length	Info
1578	8.366746	10.0.9.171	10.0.0.2	DNS	111	Standard query 0xfa17 A ip-10-0-9-171.eu-west-2.c
1579	8.366871	10.0.9.171	10.0.0.2	DNS	111	Standard query 0x91d0 AAAA ip-10-0-9-171.eu-west-
1580	8.367683	10.0.0.2	10.0.9.171	DNS	127	Standard query response 0xfa17 A ip-10-0-9-171.eu
1581	8.367796	10.0.0.2	10.0.9.171	DNS	172	Standard query response 0x91d0 AAAA ip-10-0-9-171
1690	8.464862	10.0.9.171	10.0.0.2	DNS	99	Standard query 0x2eae AAAA api.open.wisdom.alphas
1691	8.464965	10.0.9.171	10.0.0.2	DNS	99	Standard query 0x4080 A api.open.wisdom.alphasoc.
1697	8.499633	10.0.0.2	10.0.9.171	DNS	115	Standard query response 0x4080 A api.open.wisdom.
1698	8.501061	10.0.0.2	10.0.9.171	DNS	192	Standard query response 0x2eae AAAA api.open.wisd
1703	8.505463	10.0.9.171	10.0.0.2	DNS	83	Standard query 0x7481 AAAA alphasoc.com OPT
1704	8.505551	10.0.9.171	10.0.0.2	DNS	83	Standard query 0x3a03 A alphasoc.com OPT
1707	8.522861	10.0.0.2	10.0.9.171	DNS	173	Standard query response 0x7481 AAAA alphasoc.com
1708	8.526800	10.0.0.2	10.0.9.171	DNS	99	Standard query response 0x3a03 A alphasoc.com A 3
1709	8.527712	10.0.9.171	10.0.0.2	DNS	122	Standard query 0x861e TXT mlckdhokhvhtcmevvcgbggc
1906	8.701072	10.0.0.2	10.0.9.171	DNS	235	Standard query response 0x861e TXT mlckdhokhvhtcm
1907	8.728635	10.0.9.171	10.0.0.2	DNS	122	Standard query 0x8740 TXT jzxtwjwmmikyifkkigrzpio
1908	8.869328	10.0.0.2	10.0.9.171	DNS	235	Standard query response 0x8740 TXT jzxtwjwmmikyif
1909	8.929345	10.0.9.171	10.0.0.2	DNS	122	Standard query 0xbde8 TXT repusowzuzcogzgmuuvtilw

PCAP Four) Go to View > Time Display Format > UTC Date and Time of Day. What is the date and timestamp? (Format: YYYY-MM-DD HH:MM:SS) *(3 points)*

2024-05-24 10:08:50

Correct! ✔

dns

Title: Time | Type: Time (format as specified) | Fields: Enter a field ... | Oc

No.	Time	Source	Destination	Protocol	Length	Info
1578	2024-05-24 10:08:49.972483	10.0.9.171	10.0.0.2	DNS	111	Standard query 0xfa17 A ip-10-0-
1579	2024-05-24 10:08:49.972608	10.0.9.171	10.0.0.2	DNS	111	Standard query 0x91d0 AAAA ip-10
1580	2024-05-24 10:08:49.973420	10.0.0.2	10.0.9.171	DNS	127	Standard query response 0xfa17 A
1581	2024-05-24 10:08:49.973533	10.0.0.2	10.0.9.171	DNS	172	Standard query response 0x91d0 A
1690	2024-05-24 10:08:50.070599	10.0.9.171	10.0.0.2	DNS	99	Standard query 0x2eae AAAA api.o
1691	2024-05-24 10:08:50.070702	10.0.9.171	10.0.0.2	DNS	99	Standard query 0x4080 A api.open
1697	2024-05-24 10:08:50.105370	10.0.0.2	10.0.9.171	DNS	115	Standard query response 0x4080 A
1698	2024-05-24 10:08:50.106798	10.0.0.2	10.0.9.171	DNS	192	Standard query response 0x2eae A
1703	2024-05-24 10:08:50.111200	10.0.9.171	10.0.0.2	DNS	83	Standard query 0x7481 AAAA alpha
1704	2024-05-24 10:08:50.111288	10.0.9.171	10.0.0.2	DNS	83	Standard query 0x3a03 A alphasoc
1707	2024-05-24 10:08:50.128598	10.0.0.2	10.0.9.171	DNS	173	Standard query response 0x7481 A
1708	2024-05-24 10:08:50.132537	10.0.0.2	10.0.9.171	DNS	99	Standard query response 0x3a03 A
1709	2024-05-24 10:08:50.133449	10.0.9.171	10.0.0.2	DNS	122	Standard query 0x861e TXT mlckdh

PCAP Four) What is the ATT&CK subtechnique relating to this activity? (Format: TXXXX.xxx) *(1 points)*

T1071.004

Correct! ✔

dns txtquery MITRE ATT

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These are results for dns ***txt query*** MITRE ATT

Search instead for dns txtquery MITRE ATT

MITRE ATT&CK®
https://attack.mitre.org › techniques

Application Layer Protocol: DNS, Sub-technique T1071.004

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Missing: ~~ATT~~ | Show results with: ATT